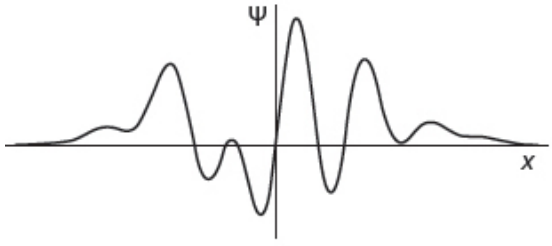
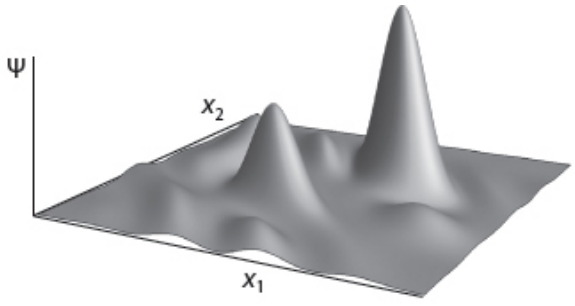
electrons and focus in on the other; the predictions we make for observations of either one can be dramatically affected by the outcome of observations of the other. The electrons are entangled.

A wave function is an assignment of a complex number, the amplitude, to each possible observational outcome, and the square of the amplitude equals the probability that we would observe that outcome were we to make that measurement. When we're talking about more than one particle, that means we assign an amplitude to every possible outcome of observing all the particles at once. If what we're observing is positions, for example, the wave function of the universe can be thought of as assigning an amplitude to every possible combination of positions for all the particles in the universe.

You might wonder whether it's possible to visualize something like that. We can do it for the simple case of a single particle that we imagine only moves along one dimension, say, an electron confined to a thin copper wire: we draw a line representing the position of the particle, and plot a function representing the amplitude for each position. (Generally we cheat even in this simple context by just plotting a real number rather than a complex number, but so be it.) For two particles confined to the same one-dimensional motion, we could draw a two-dimensional plane representing the positions of each of the two particles, and then do a three-dimensional contour plot for the wave function. Note that this isn't one particle in two-dimensional space; it's two particles, each on a one-dimensional space, so the wave function is defined on the two-dimensional plane describing both positions.



Wave function for one particle with position x



Wave function for two particles with positions x_1, x_2

Because of the finite speed of light and a finite time since the Big Bang, we can see only a finite region of the cosmos, which we label "the observable universe." There are approximately 10^{88} particles in the